TEXORA · SESSIONAL NOTES · LAB EXPERIMENT SHEET

Machine Learning and Deep Learning

Software that learns the rule from examples instead of being given the rule: a regression fitted by hand, a classifier built from class centres — and where deep learning takes over when the input is an image of cloth rather than a row of numbers.

TE-3208 · SOFTWARE AND AI IN TEXTILE MANUFACTURING SESSIONAL · EXPERIMENT 7 · LEVEL 3 TERM 2

1. Aim

To build and verify two small machine-learning models by hand — a linear regression that predicts electricity consumption from production load, and a nearest-centroid classifier that grades yarn lots — and to state correctly what deep learning adds, and what it costs, beyond these classical methods.

2. Concepts you need

- **Machine learning (ML):** a program whose behaviour is fitted to example data rather than written as fixed rules. In **supervised learning** each example carries the answer: the model learns to predict that answer (the **label**) from the measured inputs (the **features**).
- **Regression vs classification:** regression predicts a number (kWh, yarn strength); classification predicts a category (accept / reject, fault type). This lab builds one of each.
- **Training and testing:** a model is fitted on **training data** and judged on **test data it has never seen**. A model that memorises its training set — predicting it perfectly and everything else badly — is **overfitted**; only the unseen-data score is honest.
- **Linear regression** fits $\hat{y} = a + bx$ by least squares: the line that minimises the sum of squared vertical errors (residuals). It is the reference model of all prediction work — simple, fully checkable by hand, and often surprisingly hard to beat.
- **Nearest-centroid classification:** each class is summarised by its centre (the mean of its examples); a new case joins the class whose centre is nearest. It teaches the two ideas all classifiers share: a **distance** in feature space, and the need to **scale features** so that a large-unit feature (CSP in thousands) does not drown a small-unit one (U% in tens).
- **Deep learning (DL):** ML with neural networks of many layers, which learn their own features from raw input. Its textile stronghold is **images** — fabric fault detection and grading from camera pictures, where nobody can write rules for "what a slub looks like" but thousands of labelled

pictures teach the network. The costs: far more data, far more computing, and a model whose reasoning cannot be audited line by line like the regression above.

- **Where each belongs in the mill:** prediction from process numbers (energy, quality from test data, recipe outcomes) — classical ML. Judgement from pictures and signals (fault images, vibration patterns) — deep learning. Choosing the simplest model that answers the question is an engineering decision, not a fashion one.

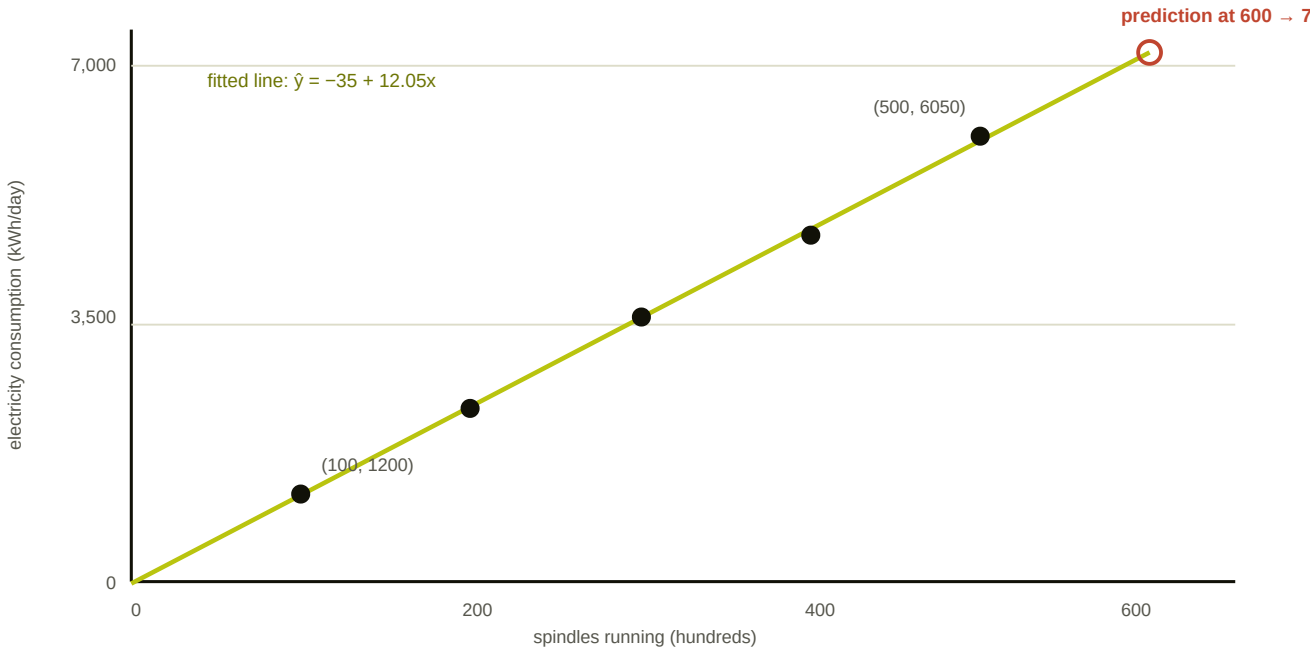


Figure 1 — The five training points of the worked example and the least-squares line fitted to them, drawn to the calculated slope (12.05) and intercept (−35). The open circle is the model's prediction for an input outside the training range — useful, but an extrapolation to be treated with care.

3. Lab procedure

1. Enter the five training records of the worked example (spindles running vs daily electricity) in the lab tool (spreadsheet or notebook).
2. Compute the regression by hand exactly as in the worked example; then obtain the tool's fitted slope and intercept and reconcile the two.
3. Plot the data and the line (Figure 1); compute the five residuals and confirm they sum to zero — a property of a correctly fitted least-squares line, and a quick fault check.
4. Build the nearest-centroid classifier of worked example B; classify the three unlabelled lots provided by the instructor by hand, then with the tool.
5. Deliberately overfit: fit a curve through the five points of step 1 exactly (a 4th-degree polynomial passes through all five) and test it at x = 600. Record how badly the "perfect" training fit behaves on new data.

6. Write the conclusion: for each of the two mill problems named by the instructor (one numerical, one image-based), state whether classical ML or deep learning is the right tool, and why.

4. Worked examples

A — Linear regression: electricity from spindles running

Five days of records from a ring-frame section (x = spindles running, in hundreds; y = electricity, kWh/day):

X	Y	X − X̄	Y − Ȳ	(X−X̄)(Y−Ȳ)	(X−X̄)²
100	1,200	−200	−2,380	476,000	40,000
200	2,350	−100	−1,230	123,000	10,000
300	3,600	0	+20	0	0
400	4,700	+100	+1,120	112,000	10,000
500	6,050	+200	+2,470	494,000	40,000
x̄ = 300	**ȳ = 3,580**			**Σ = 1,205,000**	**Σ = 100,000**

$$\text{Slope } b = 1{,}205{,}000 \div 100{,}000 = \mathbf{12.05 \text{ kWh per hundred spindles}}$$

$$\text{Intercept } a = \bar{y} - b\cdot\bar{x} = 3{,}580 - 12.05 \times 300 = 3{,}580 - 3{,}615 = \mathbf{-35}$$

Model: **ŷ = −35 + 12.05x**. Prediction for a planned day at 600 (hundred) spindles: ŷ = −35 + 12.05 × 600 = **7,195 kWh**. Residuals at the training points (actual − predicted): 1200−1170 = +30; 2350−2375 = −25; 3600−3580 = +20; 4700−4785 = −85; 6050−5990 = +60 — sum = 0 ✓. Interpretation: the slope is the marginal cost of production — every extra hundred spindles running adds ≈ 12 kWh/day; the small negative intercept is a fitting artefact of a nearly proportional process, not a physical quantity (no spindles running does not consume −35 kWh — the model is valid over the fitted range, roughly 100–600).

B — Nearest-centroid classification: grading yarn lots

Training lots, features U% and CSP (from TE-3201 testing): Acceptable — (11.5, 2350), (12.0, 2280), (11.0, 2420). Rejected — (15.5, 1950), (16.0, 1880), (15.0, 2010).

$$\text{Centroid A} = (11.5,\ 2350)\ \cdot\ \text{Centroid R} = (15.5,\ 1946.7)$$

CSP values in the thousands would crush U% in any raw distance, so scale CSP by ÷100: A = (11.5, 23.50), R = (15.5, 19.47). New lot X = (13.0, 2150) → scaled (13.0, 21.50):

$$d(X, A) = \sqrt{[(13.0-11.5)^2 + (21.50-23.50)^2]} = \sqrt{[2.25 + 4.00]} = \sqrt{6.25} = \mathbf{2.50}$$

$$d(X, R) = \sqrt{[(13.0-15.5)^2 + (21.50-19.47)^2]} = \sqrt{[6.25 + 4.12]} = \sqrt{10.37} = \mathbf{3.22}$$

X is nearer the Acceptable centre → **classify Acceptable** — and the margin (2.50 vs 3.22) is comfortably clear, unlike a case falling near the mid-line, which a sensible system flags for human review rather than deciding silently.

5. Practice tasks

1. Fit a regression by hand: operator experience x (months) = 2, 5, 8, 11 against fabric faults y (per 100 m) = 14, 11, 9, 6. Give slope, intercept, and the predicted fault rate at 14 months. ($\bar{x} = 6.5$, $\bar{y} = 10$.)
2. Using the classifier of worked example B, classify the lot (U% 15.0, CSP 1980). Show both distances.
3. A colleague trains a model on 40 dyeing batches and reports "100 % correct on the training data, 71 % on last month's production". Name the problem and state the two changes that would give an honest figure and a better model.

Answer key

1. Deviations: dx = −4.5, −1.5, +1.5, +4.5; dy = +4, +1, −1, −4. $\Sigma dx \cdot dy = (-18) + (-1.5) + (-1.5) + (-18) = -39$; $\Sigma dx^2 = 20.25 + 2.25 + 2.25 + 20.25 = 45$. Slope $b = -39 \div 45 =$ **−0.867 faults per month of experience**. Intercept $a = 10 - (-0.867 \times 6.5) = 10 + 5.63 =$ **15.63**. Prediction at 14 months: $\hat{y} = 15.63 - 0.867 \times 14 = 15.63 - 12.13 =$ **3.5 faults per 100 m**.
2. Scaled lot: (15.0, 19.80). d to A = $\sqrt{[(15.0-11.5)^2 + (19.80-23.50)^2]} = \sqrt{[12.25 + 13.69]} = \sqrt{25.94} \approx$ **5.09**. d to R = $\sqrt{[(15.0-15.5)^2 + (19.80-19.47)^2]} = \sqrt{[0.25 + 0.11]} = \sqrt{0.36} =$ **0.60**. Far nearer R → **classify Rejected**.
3. The model is **overfitted** — it has memorised the 40 training batches (including their noise) instead of learning the underlying dyeing behaviour; the 71 % on unseen production is the honest performance. Fixes: (i) judge and select models by a held-out test set (or cross-validation), never by training accuracy; (ii) simplify the model and/or train on more batches, so it is forced to learn the general pattern rather than the particulars of 40 examples.
